

AI-Based Loan Eligibility Assessment System Using Forward Chaining Algorithm

Foundation of Artificial Intelligence

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Abstract

This project implements a rule-based expert system that automates loan eligibility assessment using a Forward Chaining inference engine. Applicant facts (credit score, monthly income, existing debts, employment status, loan amount, EMI, and DTI) are validated and transformed into facts. A modular Knowledge Base of ~30+ rules is evaluated to derive deterministic outcomes: ELIGIBLE, NOT ELIGIBLE, or MANUAL REVIEW.

The system emphasizes Explainable AI: it records fired rules, shows reasoning, highlights risk factors, and provides recommendations. Ancillary features include an EMI calculator, What-If simulator, assessment history, PDF reports, and an admin dashboard. The Forward Chaining engine remains the authoritative decision-maker while an agentic workflow orchestrates validation, calculations, and explanations.

Problem Statement & Motivation



Complex, Multi-Factor Decisions

Loan assessments require synthesizing credit score, income, debts, employment, EMI affordability, tenure, and DTI — a process that is time-consuming when done manually.



Opaque ML Models

Machine learning models can predict eligibility but often lack clear, auditable explanations for decisions — a barrier for regulated finance settings.



Maintainability & Auditability

Embedding rules in code or models makes updates and audits difficult. A modular knowledge base improves traceability and governance.

Literature Survey — Landscape & Gaps

- **Existing Approaches**

Traditional manual review, credit scoring, ML/DL prediction models, and classic expert systems.

- **Advantages vs Limitations**

ML offers pattern recognition but is often a black box. Expert systems are explainable but depend on rule quality. Manual review is interpretable but slow.

- **Research Gap**

There is a need for deterministic, fast, auditable systems that combine automation with transparent reasoning and flexible, maintainable rules.





Research Objectives

Project objectives:

- Build a rule-based loan assessment system with modular Knowledge Base.
- Implement Forward Chaining inference to derive eligibility decisions.
- Calculate EMI and Debt-to-Income Ratio (DTI) and use them as facts.
- Provide Explainable AI: rules fired, decision reasons, and risk analysis.
- Include a What-If simulator, assessment history, PDF reports, and role-based access (User/Admin).
- Design an agentic workflow to orchestrate validation, calculations, and explanations.

System Architecture

Key layers:

Frontend

React.js application (Axios, React Router, Chart.js) — multi-step loan form, results, What-If UI.

Backend & API

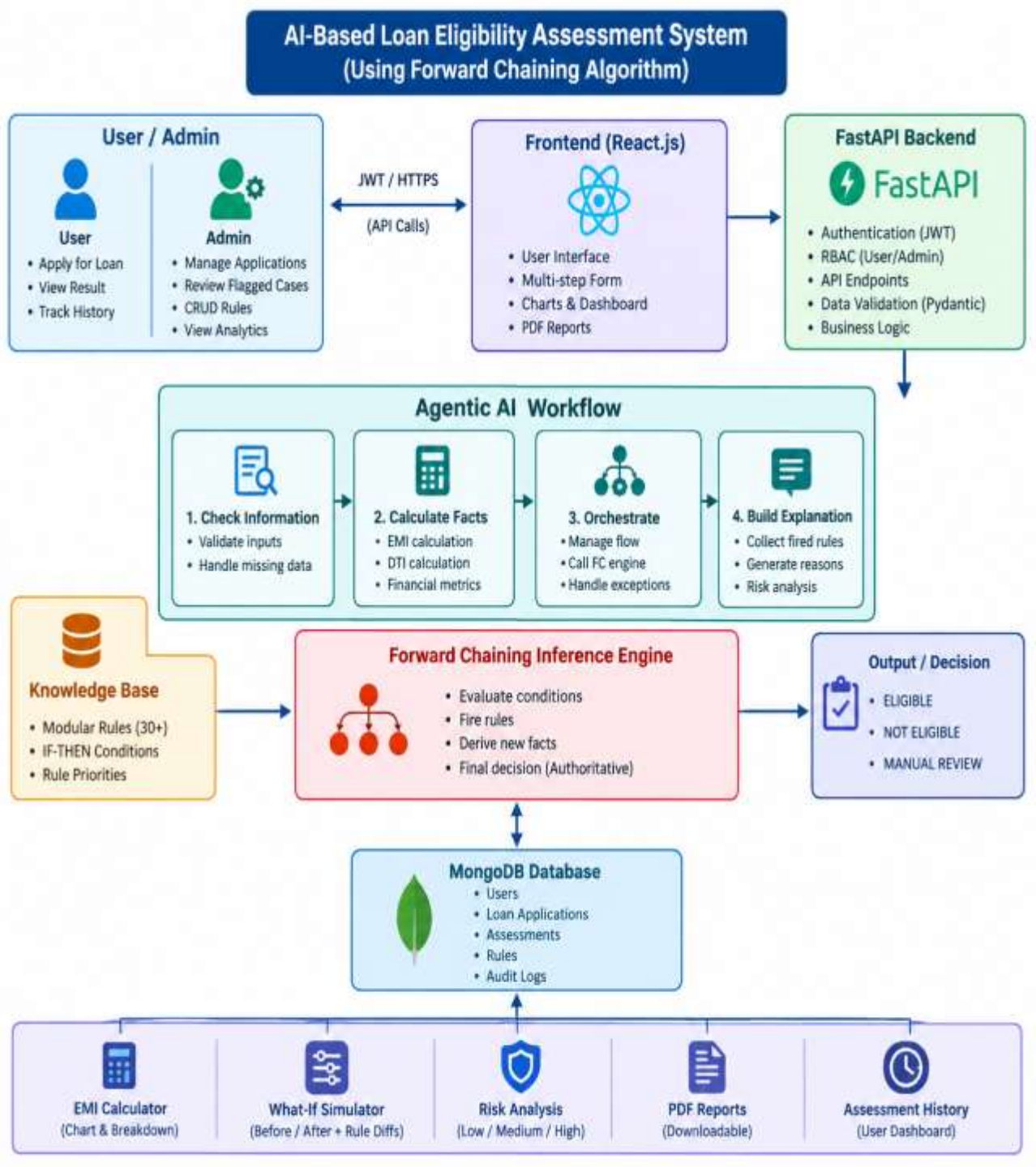
FastAPI (Python), Pydantic validation, JWT authentication, role-based access, Swagger for testing.

AI

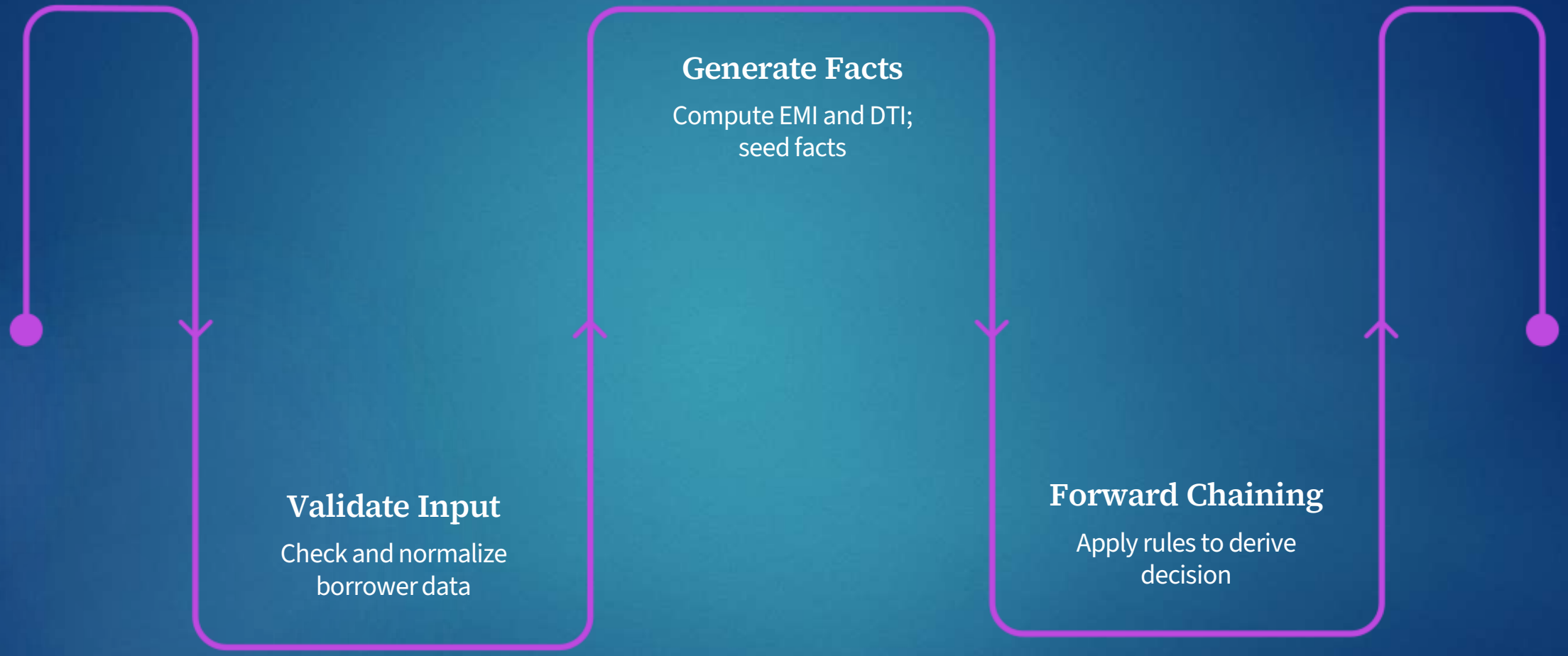
Agentic workflow for validation, EMI & DTI computation, fact generation, and orchestration of Forward Chaining inference.

DataBase

MongoDB stores users, loan applications, rules, assessment logs, and audit trails.



Forward Chaining — Algorithm & Example



Working process: validate input → compute EMI & DTI → seed initial facts → load Knowledge Base → evaluate rule conditions → fire satisfied rules (produce new facts) → repeat until no new facts → produce final decision and explanation. Example rule: IF CreditScore >= 700 AND DTI <= 40% AND Employment = Stable THEN Financially_Qualified = TRUE.



Implementation Details & Modules

Core Modules

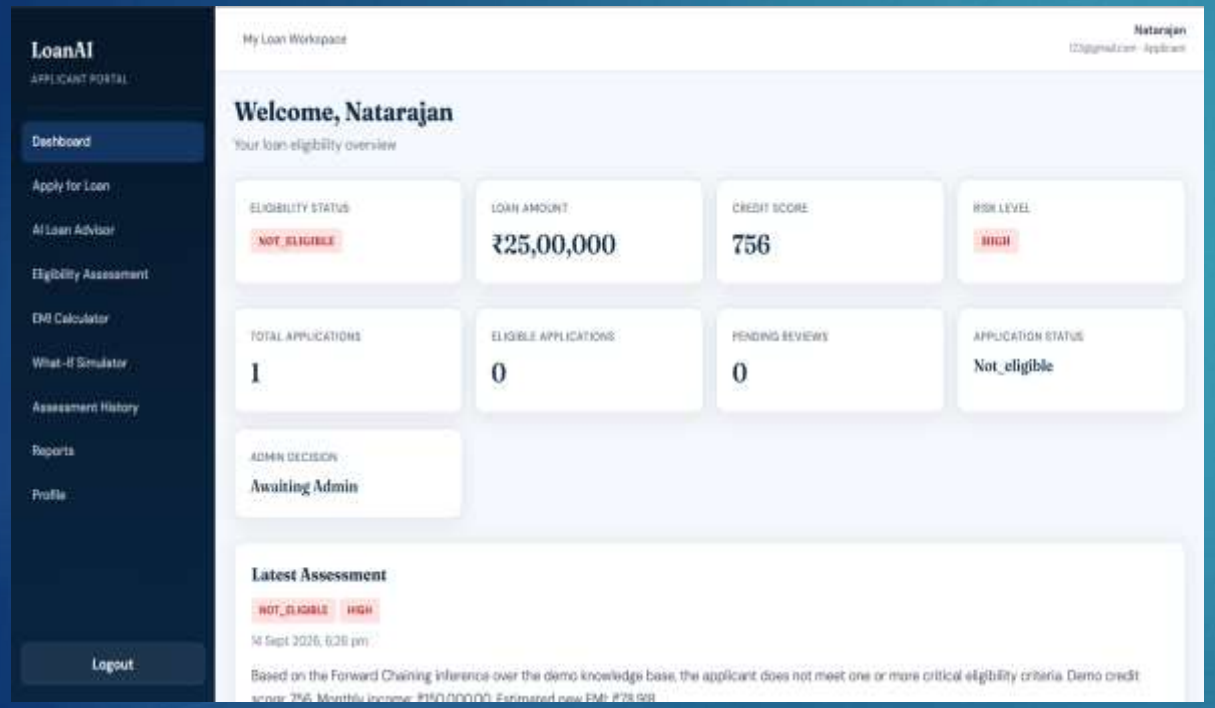
- User Module: registration, login, application submission, history
- AI Module: fact generator, Knowledge Base, Forward Chaining engine, rule firing
- Financial Module: EMI & DTI calculation, affordability checks

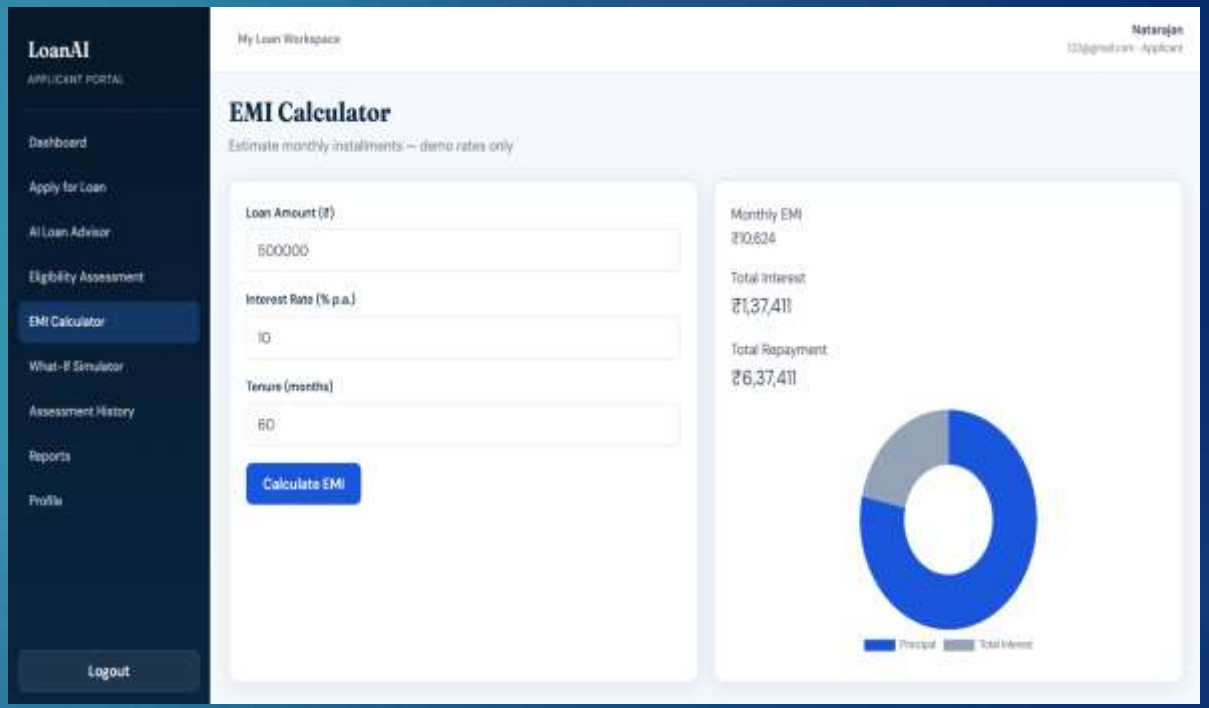
Supporting Features

- What-If simulator (recalculate and re-run rules)
- Admin: manage rules, review applications, audit logs
- Reporting: PDF export, assessment history, analytics

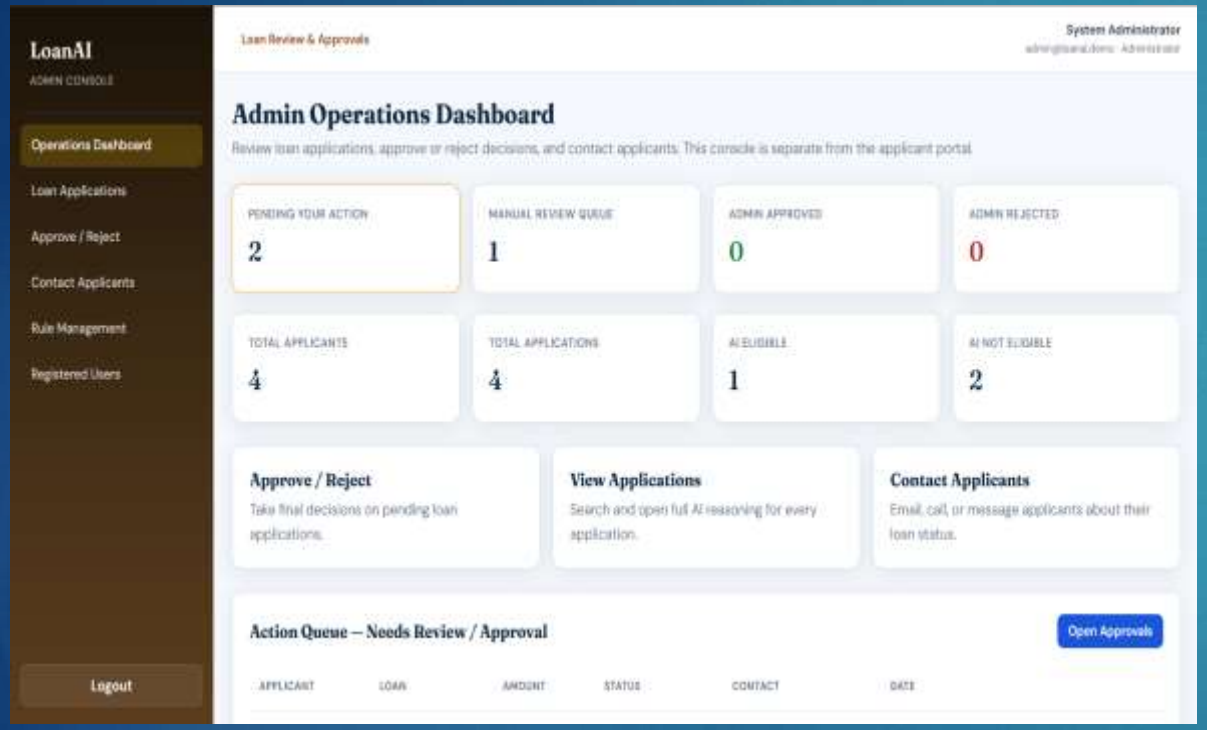
Demo Screens & User Experience

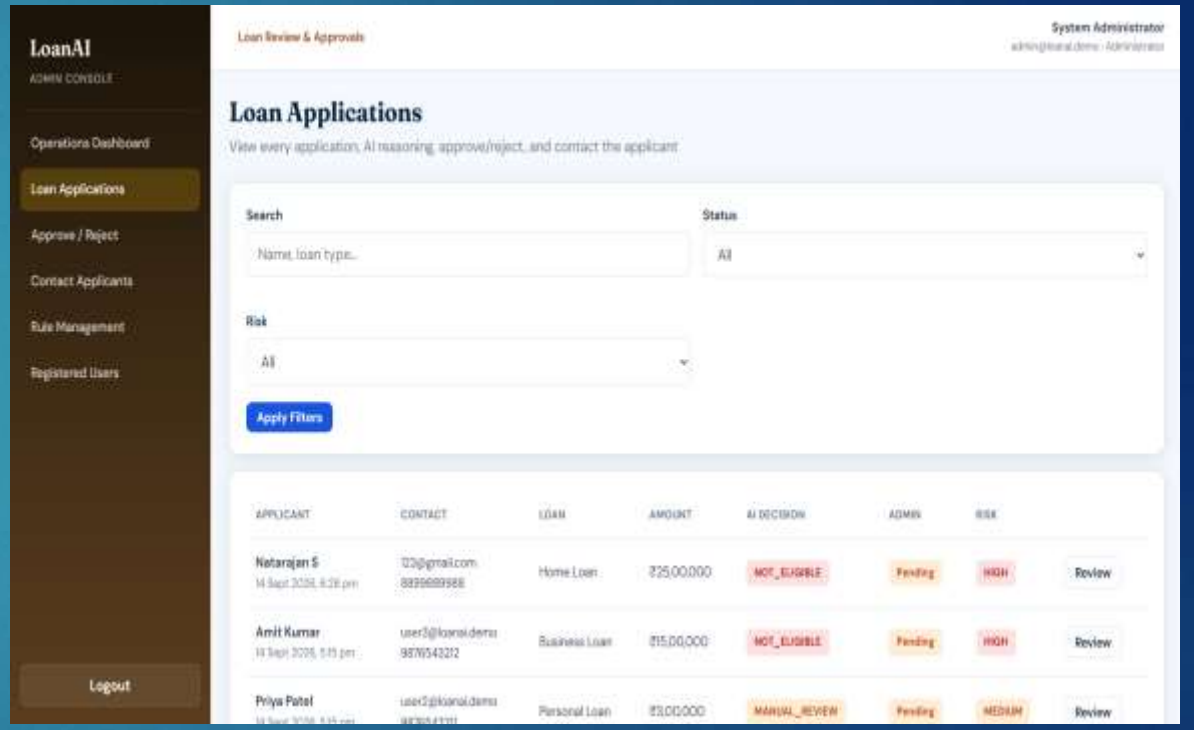
User Dashboard :





Admin Dashboard :







Results, Limitations & Next Steps

Results

Demonstrated deterministic, explainable loan decisions using Forward Chaining, EMI/DTI calculations, 30+ modular rules, What-If simulation, and role-based access.

Limitations

Rules and thresholds are illustrative for academic purposes and are not production underwriting criteria. System behavior depends on rule completeness and quality; maintenance processes are essential.

Next Steps

Expand rule authoring UI, add versioned rule testing, hybridize with probabilistic scoring for borderline cases, integrate secure audit pipelines, and run user studies with domain experts to refine thresholds and workflows.